

# Satish Kumar Jakkula

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## Professional Background

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Data Scientist with 3 years of experience in **Machine Learning, Natural Language Processing, Deep Learning, Generative AI, and Time Series**. Proficient in **Python, Scikit-Learn, and TensorFlow** for building predictive models, with expertise in EDA, data preprocessing, and feature engineering. Skilled in developing and deploying end-to-end Artificial Intelligence solutions using Flask, and cloud platforms like **Microsoft Azure**.

## Primary Skills

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### Technical Skills

- Machine Learning (ML)
- Deep Learning (DL)
- Natural Language Processing (NLP)
- Generative Artificial Intelligence (Gen AI)
- Time Series Forecasting
- Exploratory Data Analysis (EDA), Data Preprocessing and Feature Engineering
- Predictive Analytics
- Clustering

### Programming Languages, Frameworks, and Libraries

- Python
- TensorFlow, Scikit-Learn, nltk, spacy, Numpy, Pandas, Open AI, pytorch(beginner)
- Flask, Rest API's

### Development Tools and Environments

- Linux, Putty, WinSCP
- Visual Studio Code (VS Code), Spyder, Eclipse, Jupyter Notebook, Anaconda
- GitLab, Postman
- MS Excel (pivoting, Regression Analysis, etc.)

## Additional Skills

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- Prompt Engineering.
- Automation and Scripting (e.g. Python Scripting)
- Microsoft Azure.
- Agile Methodologies.
- Soft Skills: Problem-Solving, Critical Thinking, and Collaboration

## Professional Summary

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- **3 years of experience** as a Data Scientist, specializing in **Machine Learning (ML), Natural Language Processing (NLP), Time Series Forecasting**, and **Generative AI**.
- Explored and analyzed **20+ datasets** over 3 years, gaining expertise in **data distributions** (Gaussian, Poisson, Uniform) and **hypothesis testing** (Z-score, T-score, null and alternative hypothesis).
- Conducted **Exploratory Data Analysis (EDA)**, and visualization using **Plotly, Matplotlib, and Seaborn**.
- Proficient in **feature engineering** (encoding, scaling techniques: **Standard Scaler, Min-Max**; transformation techniques: **log10, Box-Cox, square, square root**) and **feature selection** (**recursive feature elimination, Lasso(L1), forward/backward selection**).

- Handled **class imbalance** using **SMOTE**, **SMOTETomek**, and **random over/under sampling**, and performed **cross-validation** (**K-Fold**, **Stratified K-Fold**) to ensure robust model performance.
- Developed and deployed **end-to-end ML solutions** for diverse use cases, including **employee attrition**, **bank customer churn prediction**, **sentiment analysis**, **ticket classification**, **UNSPSC**, and **server capacity forecasting**.
- Expertise in **supervised and unsupervised learning algorithms** such as **Random Forest**, **Logistic Regression**, **XGBoost**, **LightGBM**, **KNN**, **Naive Bayes**, **SVC**, **K-Means**, and **LDA**.
- Improved model performance by **15% accuracy** and reduced runtime by **20 minutes** through **GridSearchCV** and **RandomSearchCV** for hyperparameter tuning and parallel processing with **n\_jobs**.
- Built and optimized **NLP pipelines** using **text cleaning** (regular expressions, stop words removal, tokenization, lemmatization), **vectorization** (TF-IDF, Bag of Words, Word2Vec, Sentence Transformers - all-MiniLM-L6-V2), tokenizer and padding, along with **NER**, **parser**, and **tagger**, achieving **87% accuracy for sentiment analysis**, **85% accuracy for ticket classification**, and **99% for Major Incident prediction**.
- Implemented **clustering solutions** using **K-Means**, **DBSCAN**, and **Hierarchical Clustering**, leveraging **Sentence Transformers** for semantic understanding and **Elbow Method** and **Silhouette Score** for optimal cluster selection.
- Integrated **Generative AI solutions** using **Azure OpenAI**, reducing costs by combining **K-Means clustering with prompt engineering** for concise cluster naming.
- Utilized **pretrained models** like **XLM-RoBERTa from Hugging Face** for sentiment analysis and **Azure Speech-to-Text** for multilingual audio transcription.
- Expertise in **time series forecasting** using **ARIMA**, **SARIMAX**, **Prophet**, **N-BEATS**, and **Exponential Smoothing**. Skilled in handling **trend**, **seasonality**, **residuals**, managing **missing values** (imputation via mean, median, linear/spline interpolation), and addressing **missing dates**. Proficient in **outlier detection** using **IQR**, **MAD**, **standard deviation**, and **percentile methods**. Achieved **MAPE < 15%** for deterministic trend-seasonal data and **MAPE < 20%** for stochastic trend-seasonal data.
- **Deep Learning Expertise**: Proficient in tuning **batch size**, **epochs**, and **learning rates** for optimal performance, implementing **optimizers** (ADAM, SGD) and **activation functions** (Sigmoid, TanH, ReLU), and using **dropout** and **pooling layers** to prevent overfitting and reduce computational complexity.
- **Deep Learning Architectures**: Hands-on experience with **RNNs**, **LSTMs**, and **GANs** for tasks like **sequential data analysis**, **image processing**, and **synthetic data generation**, including addressing **vanishing gradients** and implementing **forward/backward propagation**.
- Evaluated models using **classification metrics** (accuracy, precision, recall, AUC score, specificity, sensitivity) and **regression metrics** (MAPE, MAE, MSE, RMSE, SMAPE), addressing **high variance and high bias** scenarios.
- Designed and deployed **asynchronous workflows** using **Redis** as a messaging broker and **Celery** for task queue management.
- Proficient in using **Git**.
- Proficient in **cloud-based development and deployment** using **Azure Virtual Machines**, **Azure ML Studio**, and **Linux servers** (Putty, WinSCP).

## Experience

### Project - 1: TCS Cognix™ Rapid AI (Current)

November 2022 – Present

**Client:** Tata Consultancy Services

**Role:** Data Scientist

#### Roles and Responsibilities:

- Developed and maintained **Jupyter notebooks** for various datasets, streamlining **data preprocessing**, **model development**, and **evaluation**.
- Designed and deployed **AI solutions** tailored to business needs, ensuring automation and data-driven decision-making.
- Optimized **AI workflows** by implementing efficient **data handling**, **feature engineering**, and **model tuning** techniques.

- **Integrated multiple ML and DL algorithms** into the platform, improving prediction accuracy and clustering capabilities.
- **Developed time series forecasting solutions** to predict trend in business data.
- **Worked on speech-to-text transcription and audio data analysis**, enabling **multilingual processing**.
- **Conducted rigorous testing and Validated models** using appropriate **evaluation metrics** to ensure reliability and scalability.
- **Documented processes and findings**, ensuring reproducibility and knowledge sharing.
- **Collaborated with cross-functional teams** to translate business problems into AI-driven solutions.

## **Project - 2: Machine First Delivery Model™ (MFDM™).AI**

March 2022 – October 2022

**Client:** Tata Consultancy Services

**Role:** AIML Developer

### **Roles and Responsibilities:**

- **Developed multiple Jupyter notebooks** for various use cases, including **Topic Modeling and Sentiment Analysis**.
- **Built predictive models** for app installation through ads using **machine learning algorithms**.
- **Developed AI solutions** for **employee attrition prediction** and **bank customer churn analysis**.
- **Enhanced pre-existing notebooks**, making them ready for **seamless integration with applications** for business users.
- **Deployed machine learning models** as **Flask APIs**, enabling real-time predictions and accessibility.

## **Education**

**Brilliant Institute Of Engineering And Technology**, B.Tech in Electrical and Electronics Engineering

July 2017 – September 2020

- CGPA: 7.2

## **Achievements**

- **Promoted to Systems Engineer** after successfully **cracking the Elevate Wings exam** as part of TCS internal assessment.
- **Received the Special Initiative Award** for winning the **InfoSecurity Fest** conducted at the Unit level.
- **Honored with the Best Mentor Award, Badge, and Special Lanyard** for **mentoring associates** to complete the **AI Practitioner certification**.
- **Awarded by GEMS** for multiple achievements, including the **Xcelerate Warrior Certificate** and **MFDM AI Practitioner Certification**.

## **Certifications**

- **Oracle Cloud Infrastructure Gen AI Professional – 2024 Certification.**
- **Azure Data Scientist (DP-100) Certification.**
- **MFDM AI Practitioner Intermediate – TCS Certification.**

## **Declaration**

I hereby declare that the information provided above is true and accurate to the best of my knowledge.